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Variables Correlated with Retention in High Churn Risk Groups in the Telecommunications Industry

To create the initial dataframe, I removed columns relating to customer ID, survey responses, and other miscellaneous information that is better captured in different variables; for example, latitude and longitude are not very helpful if the customer's state of residence is already given. I felt that the variables I omitted did not offer much benefit and could reasonably be pruned to better the results of the analysis. A further description of each variable can be found in the attached data dictionary file.

To prepare the data for further analysis, I standardized all of the quantitative variables, converted the yes/no values to 1s and 0s respectively, and one hot encoded the other categorical variables.

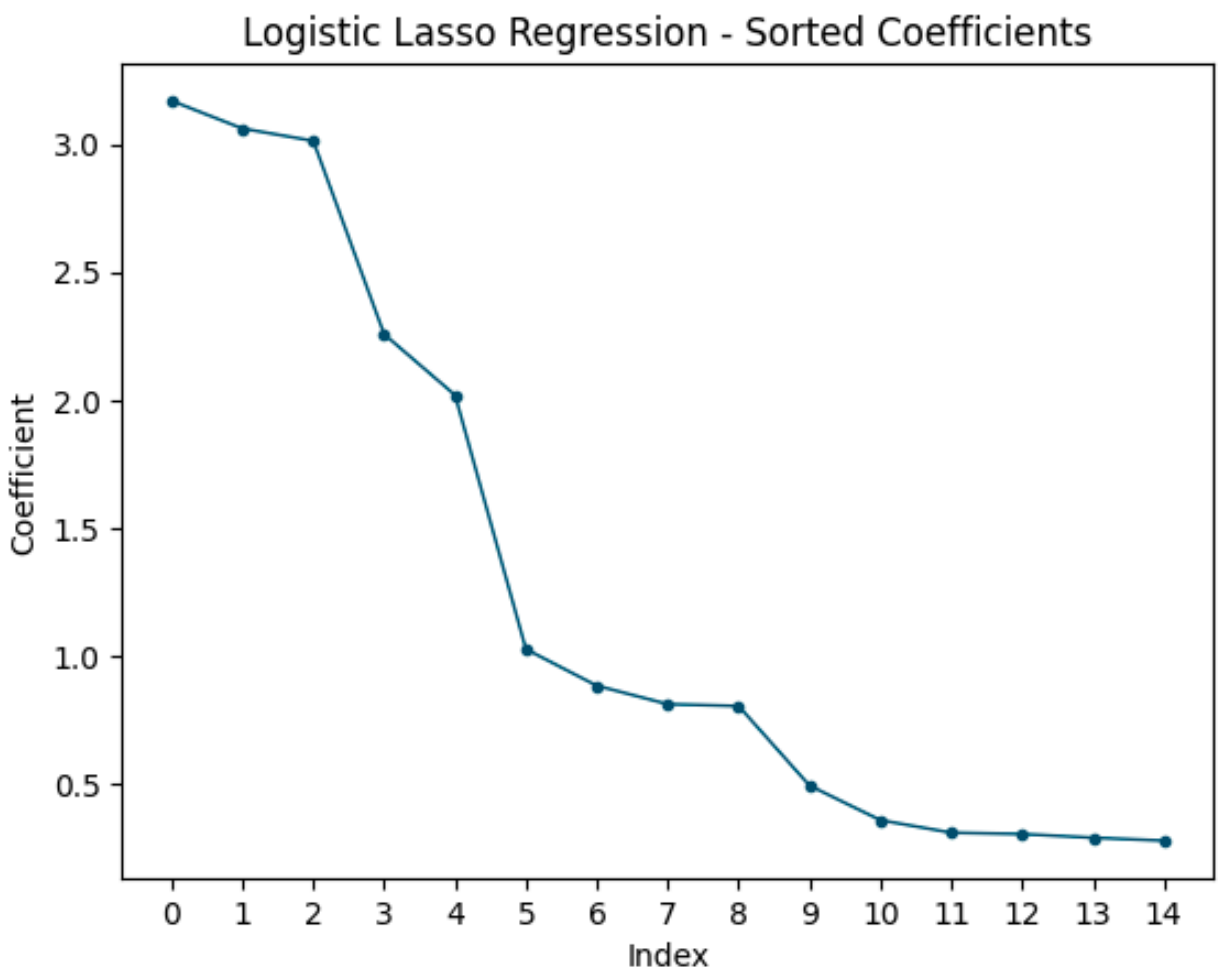
My first analysis is to determine which factors are correlated with high churn rates. By employing three different techniques for feature selection, I should be able to narrow my list of significant variables considerably. Given that I have 30 independent variables, and one dependent, categorical variable, I decided to use these techniques:

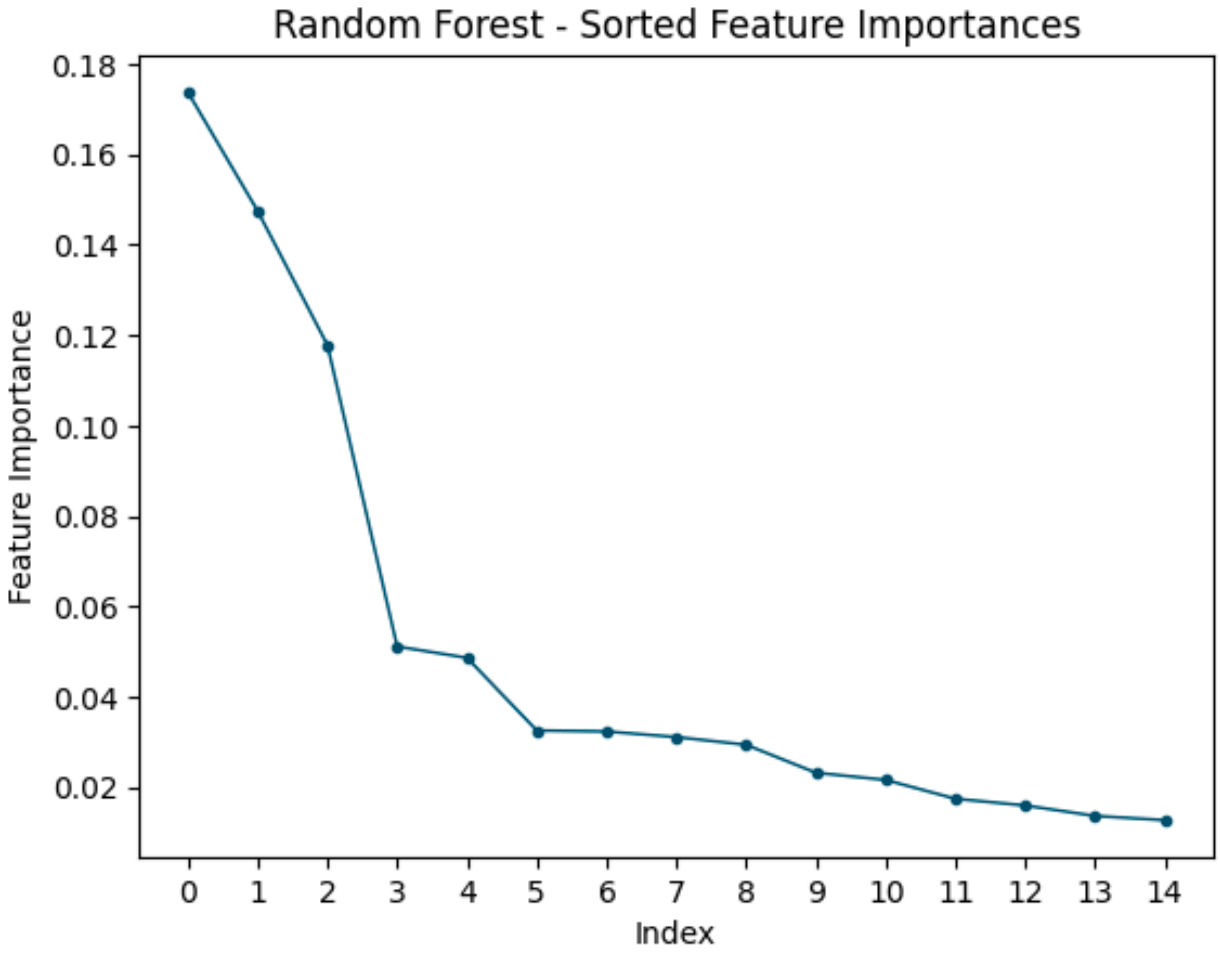
1. Logistic lasso regression
2. Random forest classifier
3. Gradient boosting classifier

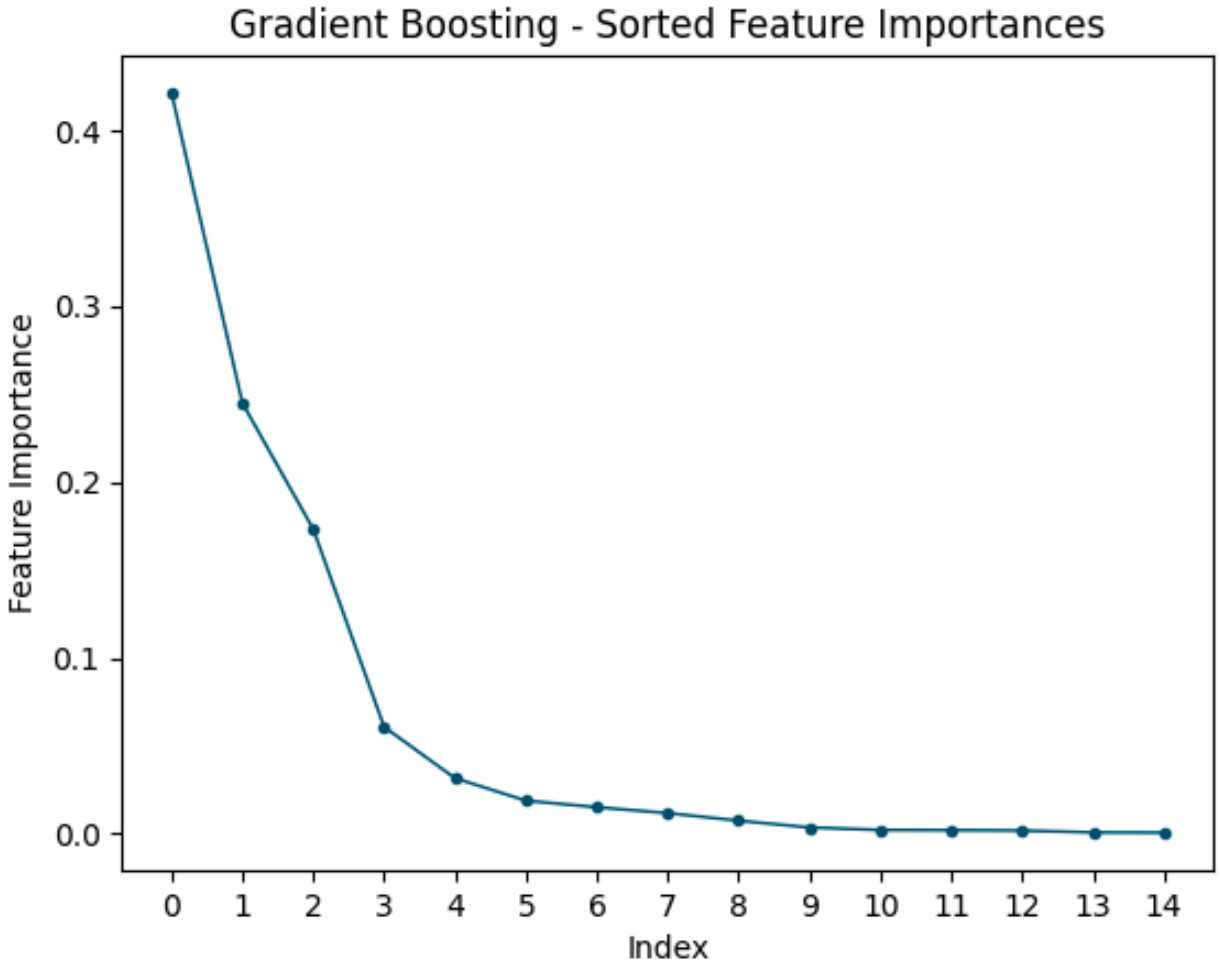
Despite train-test-splits being common for these methods, I find that it is not necessary for this instance. I am employing these as a means of feature selection and I

am not trying to build a predictive model, so I will only be splitting the data by the dependent and independent variables for the modeling phase.

Once I had fit the models to the data, I extracted and sorted the coefficients from the logistic lasso regression, and the feature importance values from the random forest and gradient boosting classifiers. I then plot the top 15 values and visually examine where the high coefficient/feature importance grouping drops off for each method. Then I use their index to create a list of all of the highly correlated values from each model and then filter by which ones were significant in all three models.



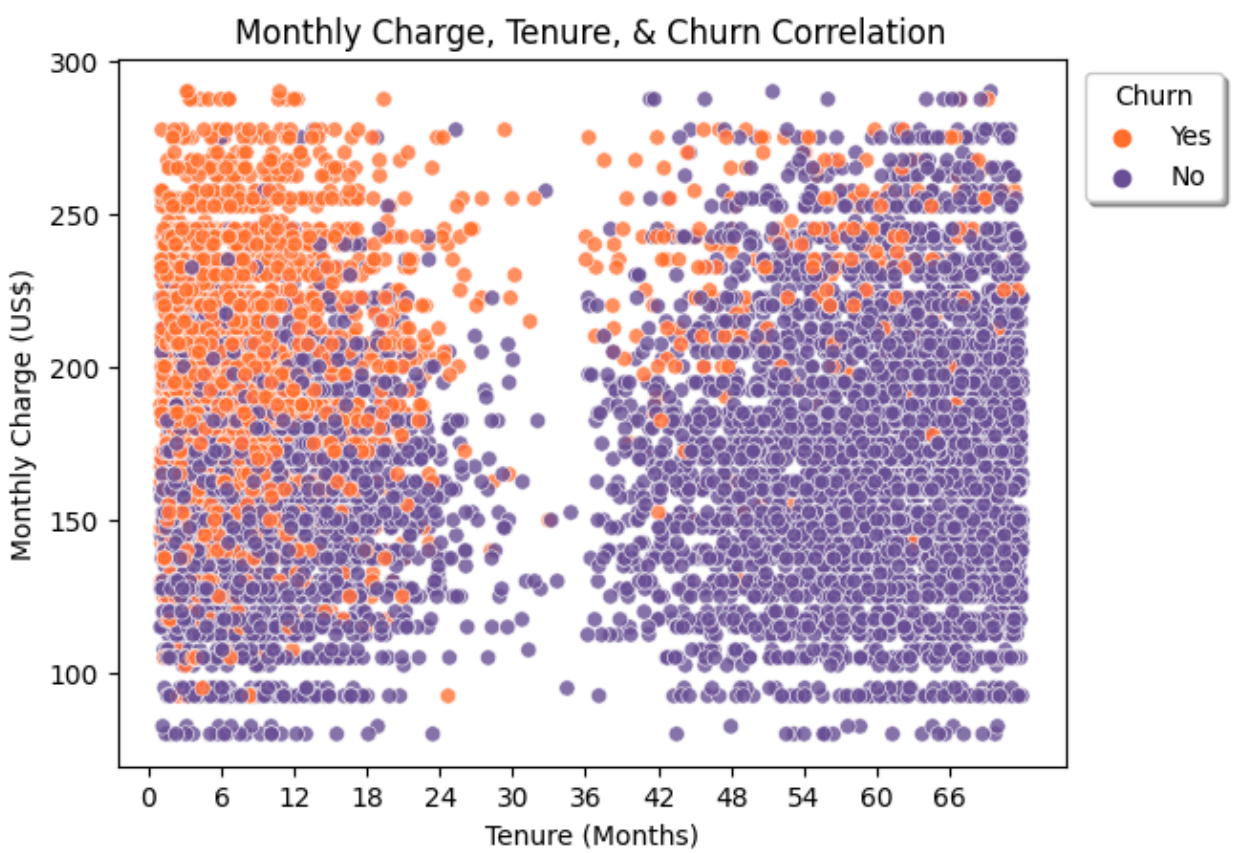

Random Forest - Sorted Feature Importances
Feature Importance
0.18
0.16
0.14
0.12
0.10
0.08
0.06
0.04
0.02
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14
Index


Gradient Boosting - Sorted Feature Importances
Feature Importance
0.4
0.3
0.2
0.1
0.0
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14
Index

The logistic lasso regression had five variables in the highly correlated grouping, while the random forest and gradient boosting classifiers only had three. Between the three models, there were only two variables that were significant in all three:

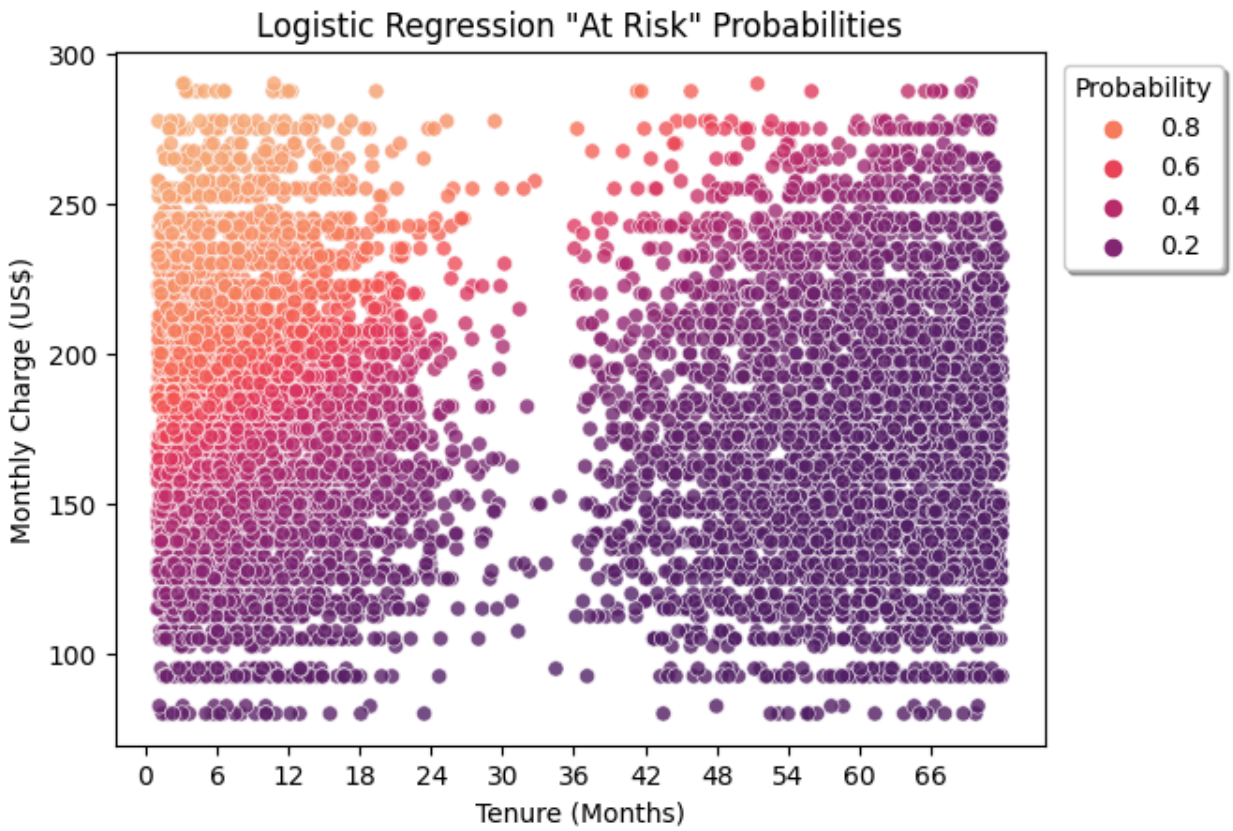
1. Tenure (Months)
2. Monthly Charge (US$)

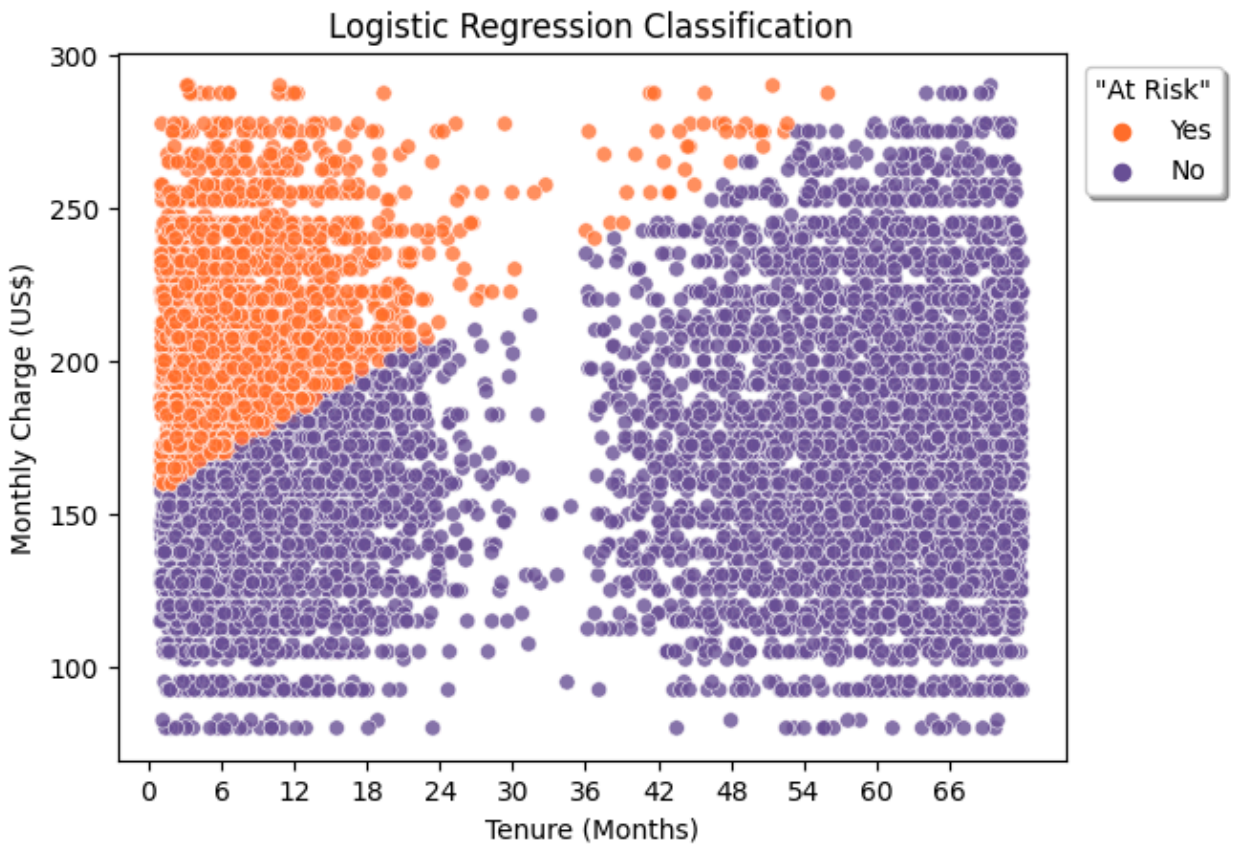
Lastly, to confirm the results, I plotted churn (yes/no) as a function of tenure and monthly charge on a scatterplot, color-coded by churn. From the graph, we can see that newer customers with a higher monthly payment are correlated with higher churn rates.



After looking at the graph up above, I became curious as to what the blue dots in the upper left corner had in common. It appears that the shorter a customer's tenure with this telecommunications company and the higher their monthly payment is, the more "at risk" they are for terminating their services. However, despite their high risk, many customers in this grouping still decide to stay. My next focus, and second analysis, became identifying the variables most associated with customer retention in high churn risk groups.

To move forward with this portion of the analysis, I decided to define a subset of customers in the dataset: the "At Risk" group. This is defined as having a sufficiently high monthly charge as well as being a sufficiently new customer. To determine where this cutoff would fall, I used logistic regression. The first graph shows the logistic regression probabilities for a customer's termination of services, the second shows where the logistic regression predicted termination of services. If the logistic regression predicted yes, they were assigned to the "At Risk" group. Both of the predictions and probabilities were stored as new values in the dataframe.


Logistic Regression "At Risk" Probabilities
Monthly Charge (US$)
Tenure (Months)
Probability
0.8
0.6
0.4
0.2


Logistic Regression Classification
Monthly Charge (US$)
Tenure (Months)
"At Risk"
Yes
No

Now that I have the "At Risk" subset of my data, I needed to find which variables were significantly different between the customers that actually terminated their services, and those that did not, but were predicted to have done so. This analysis flips the first on its head by making churn the independent variable instead of the dependent variable. As such, each variable will be need to be tested individually using one of two tests:

1. Mann-Whitney U test for the quantitative variables
2. Chi square test for the categorical variables

The reason I chose a Mann-Whitney U test over a traditional two-sample t-test is because many of my data are not normally distributed and some of them are ordinally measured. This can negatively impact a t-test, but a non-parametric approach remains reliable.

My significance level has been set to 0.05. With a Bonferroni correction, the significance level drops to 0.0167. This eliminates 4 of the 10 significant variables identified by my tests, but I will mention which ones those are after the data visualization. The ten significant variables are:

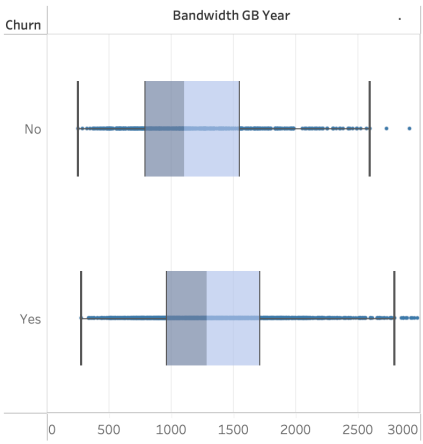
* Yearly equipment failure
* Bandwidth (GB/Year)
* Techie
* Contract
* Internet service
* Online security
* Online backup

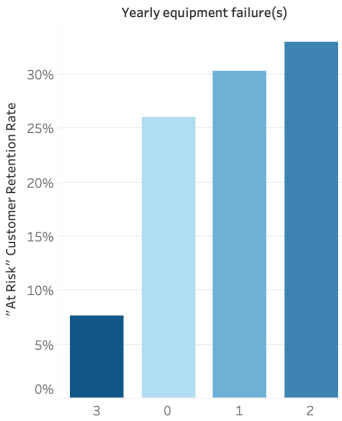
* Streaming TV

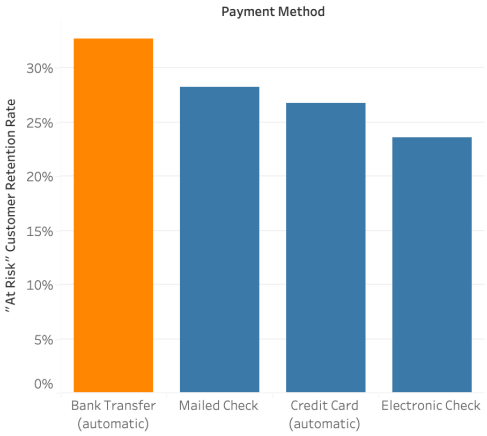
* Streaming movies

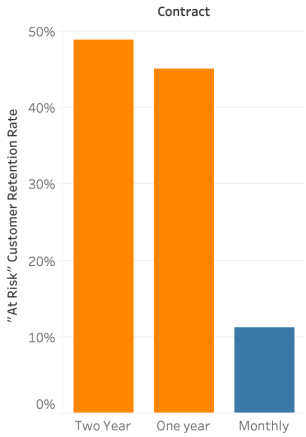
* Payment method

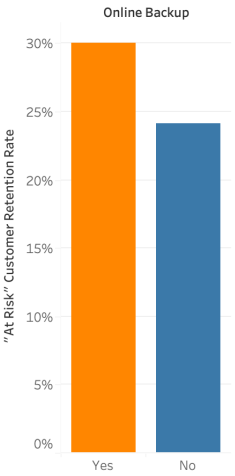
In order to more fully understand how these variables interact with churn, I uploaded the dataframe to tableau and created the following dashboard.

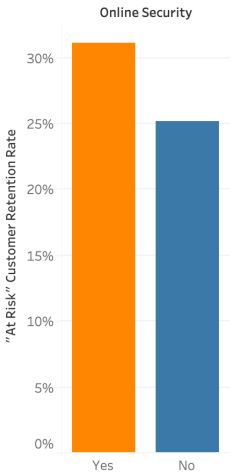


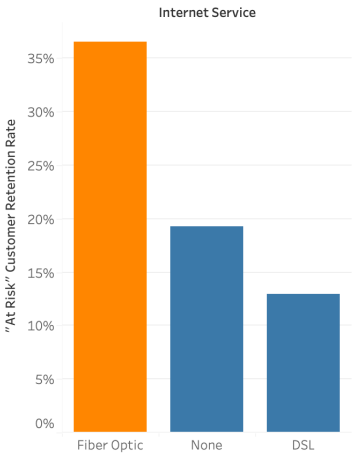


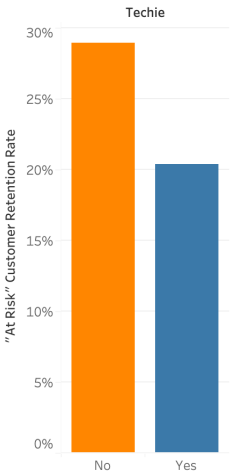


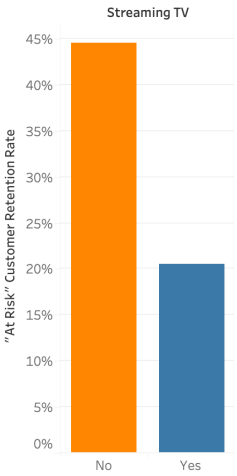


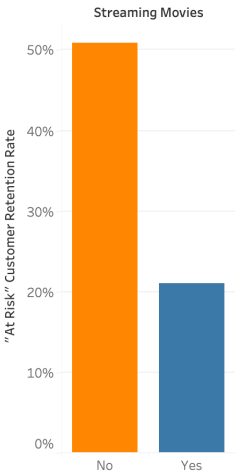












The Y-axis shows the retention rate as a percent of "at risk" customers; the higher the percent, the greater the retention rate. The following correlations have a positive relationship with churn:

* Bandwidth usage (GB/year)
* 3+ yearly equipment failures
* Streaming TV services
* Streaming movie services
* Someone considering themselves a "techie"

The following have a negative relationship with churn:

* Automatic bank transfers as payment method
* Yearly contracts
* Online backup
* Online security add-on
* Fiber optic internet

The bonferroni correction removes the following variables from the list of significance:

* Yearly equipment failure
* Payment method
* Online backup
* Online security add-on

Looking at the chart, this makes sense because these were the four with the most even distributions. That being said, as I have looked for the narrative that the data

are saying, I do not believe that the potential lack of significance of these variables changes anything.

Conclusion

The variables with a positive correlation to churn and the variables with a negative correlation to churn tell the same story: "At risk" customers who bundle their telecommunications services in yearly contracts and pay with automatic bank transfers are correlated with higher retention rates. Additionally, bundling streaming services for TV shows and movies were correlated with lower retention rates.

While the streaming services seems a little odd, everything else seems to be reasonable. Yearly contracts deter customers from terminating their services too quickly, lowering the churn rate. Automatic bank transfers make paying bills easier, making it a better and smoother experience for the customer. Additionally, the more services a customer receives from the telecommunications company, such as online security, backup, and fiber optic internet, the more justified a higher monthly charge will seem to them. Lastly, customers who consider themselves techies versus those who do not probably have a large gap in technical knowledge. It may be an easier decision for a 24 year old adult to change their telecommunications provider than for an 86 year old senior citizen.

Recommended Course of Action:

There are three courses of action that I would recommend that this telecommunications company try:

1. Start a promotional campaign that bundles multiple services for those starting a yearly contract.
2. Offer a one time incentive for customers to set up an automatic bank transfer.
3. Try reducing streaming service packages for a period of time to see if retention rates change for at risk groups and not at risk groups alike.
4. Workshop some ideas for "easy set up" and "easy use" telecommunications services to attract non-techie customers.

As with any analysis rooted in correlation, correlation is not causation. These are ideas that may prove beneficial to the business, but continue to monitor churn rates throughout any of these courses of action to ensure that they are having a positive effect.

Further Analysis Ideas:

During my analysis, there were plenty of features in the data that I would have loved to explore, but either did not have the time, or it was not directly applicable to this project. I would like to present one of those here that might be useful at a later time.

While charting churn as a function of monthly charge and tenure, I noticed that there is a gap in tenure from approximately 24 to 36 months.

* Why is there such low customer representation from 1-1.5 years ago?

* Do the factors correlated with high churn rates differ between newer and long-term customers?

* Do the factors that contribute to better retention rates differ between newer and long-term customers who are at risk for high churn rates?